DATA / MSAI / MSML 641 · GROUP 6 · FINAL PROJECT · AUGUST 19, 2026

AdInsight

Shows you how an ad is trying to persuade you — in plain language, with the evidence to check it.

DETECTS SEVEN PERSUASION TACTICS

Urgency · Scarcity · FOMO · Fear appeals · Authority · Social proof · Exaggerated claims

THE TEAM

Chris Pedretti
Engineering · Data & Eval

Shashank Ashoka
Architecture · OCR/VLM · Agents

Ciara Cameron
Review Insights · Evidence

Jonathan Kim
App · Browser Extension

AREA 1 · PROBLEM AND USERS

Persuasion you can't see, aimed at the people who pay most for it

WHO IT'S FOR

Adults 60 and older — heavily targeted by health, supplement, insurance, and financial ads on Facebook, Instagram, YouTube, and news sites.

$2.4B
FTC-reported fraud losses, ages 60+, 2024

4×
growth since 2020 (~$600M) — highest median loss of any age group

WHAT THEY DID BEFORE ADINSIGHT

Our interviews: ignore it, click through and judge the landing page, Google it, check Amazon or Reddit, ask family. Older users verify socially or buy first and judge after. Manual, inconsistent, and often too late.

THE TACTICS WE NAME

Urgency · Scarcity · FOMO · Fear appeals · Authority · Social proof · Exaggerated claims

WHAT ADINSIGHT DOES INSTEAD

- **Names the tactic** — urgency, fake scarcity, authority framing — at the moment the user sees the ad.
- **Explains in plain language**, quoting the ad's own words back as evidence.
- **Educational, not judgmental** — no "scam / not scam", no legal claims. It gives a reason to slow down.

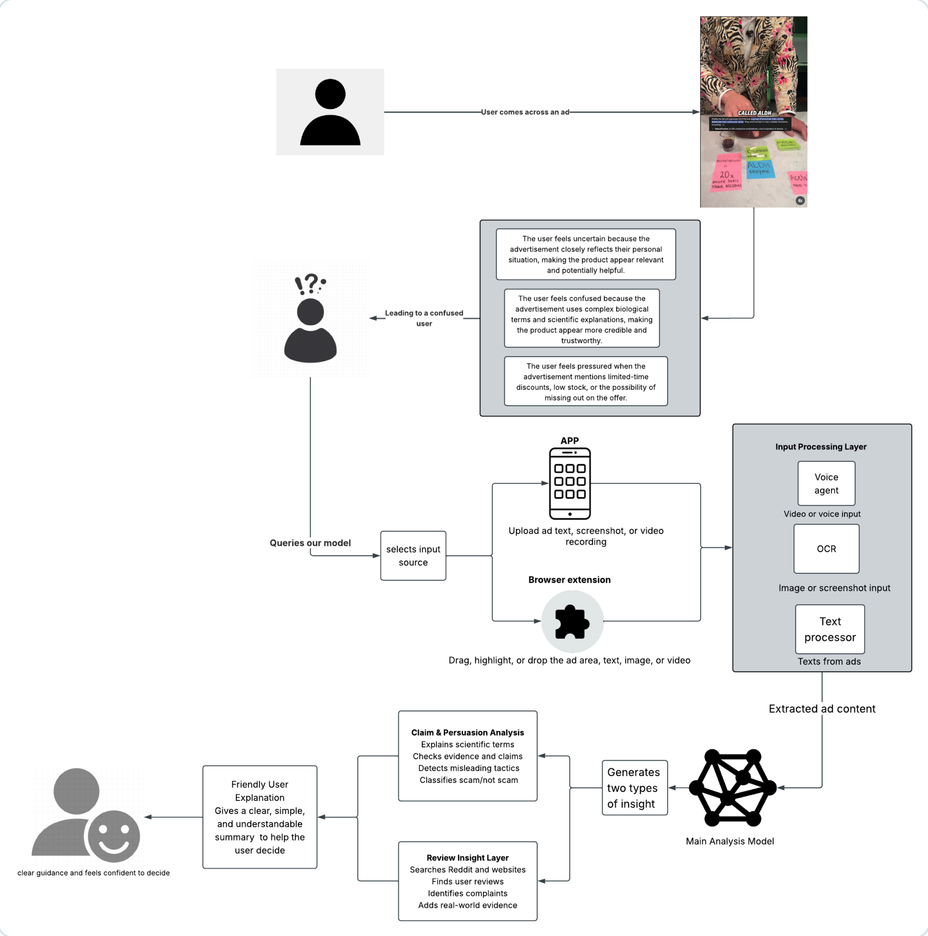
Positioning: the concern is rarely a false statement — it's the framing. "Only 3 left." "Doctor recommended." "Offer ends tonight."

AREA 2 · PRODUCT

One analysis engine, three ways in

- **Three surfaces, one backend.** Web app, desktop window, and a Chrome extension all POST to the same Flask `/api/analyze`.
- **Read.** Image ads: VLM transcription (transcription-only prompt — it may not name tactics or judge) with local RapidOCR + spacing repair as offline fallback. The extension shoots 3-frame bursts to catch rotating creatives.
- **Classify.** Fine-tuned DistilBERT, 8 classes; long ads scored in sliding windows so a tactic at the end still counts.
- **Explain.** Every finding is badged with its evidence: **model pick** (with confidence) or **phrase found** (traceable to our labeling guidelines). Below 60% confidence with no phrase, it says "not sure" instead of guessing.
- **Check.** Review layer: independent-source web search with advertiser-owned domains excluded and affiliate content flagged; falls back to clearly-badged sample data without API keys.

Engineering: 183 automated tests (126 Python + 57 JS) · model ships as 134 MB float16.



System architecture — ad in, plain-language explanation out

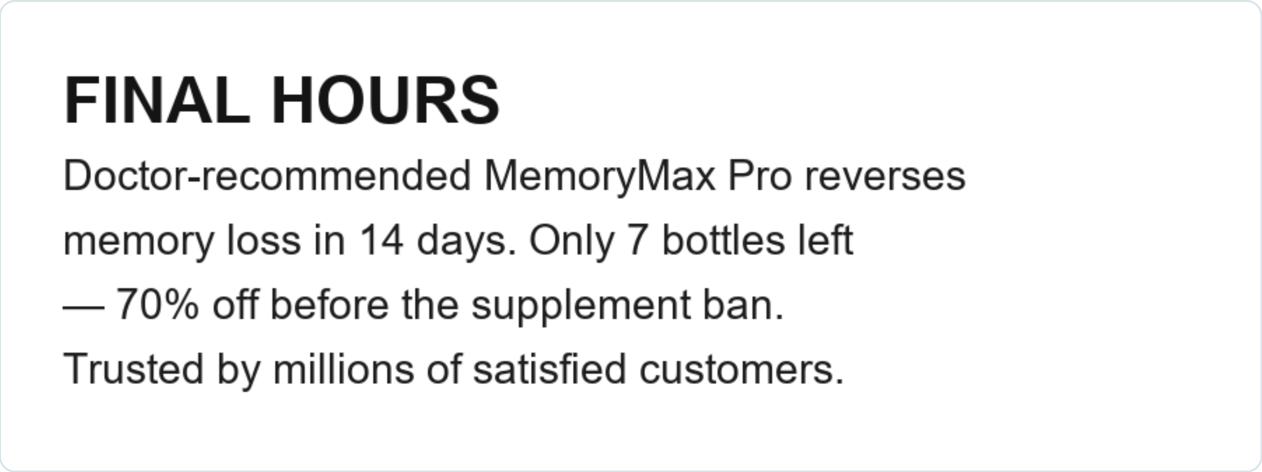
Watch it work

1. **Paste a loaded ad.** "FINAL HOURS: Doctor-recommended MemoryMax Pro… only 7 bottles left" → five tactics, each quoting the ad, plus the banner: *"Take your time. Nothing requires a decision today."*

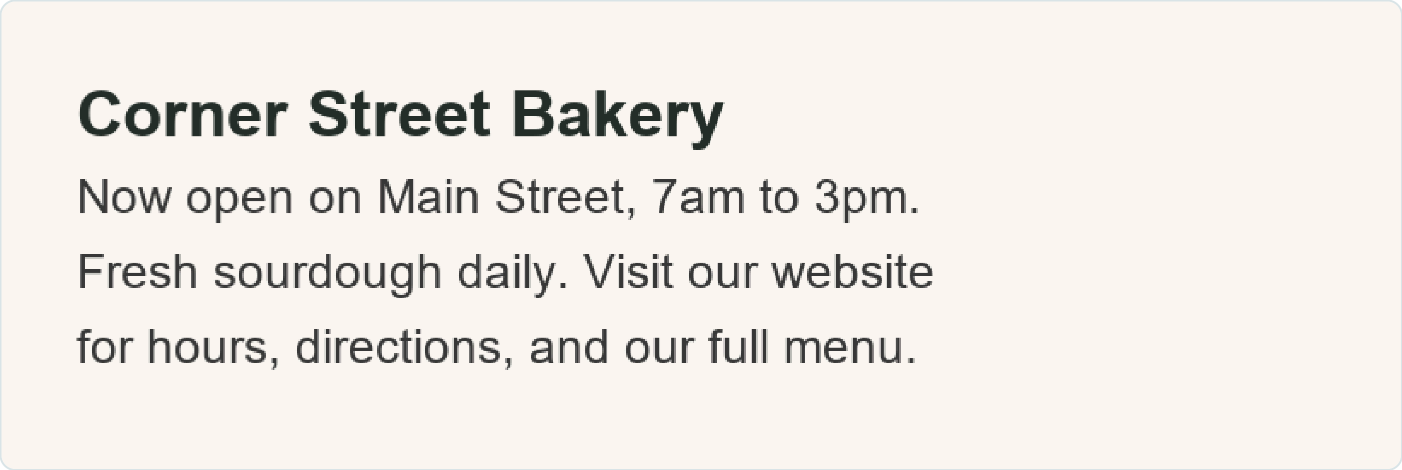
2. **Upload a screenshot.** The supplement ad image → *"What we read from your image"* (OCR ~98% confidence, spacing repaired) → six tactics. A bad scan is visible, never silent.

3. **Try to fool it.** A clean bakery ad → **Neutral** — no false alarm. (It used to cry "Social Proof, 75.8%" before we fixed the dataset.)

4. **Meet it where ads live.** The Chrome extension overlays a real page: hover an ad, or screenshot-capture a video ad in place.



The loaded ad — six tactics



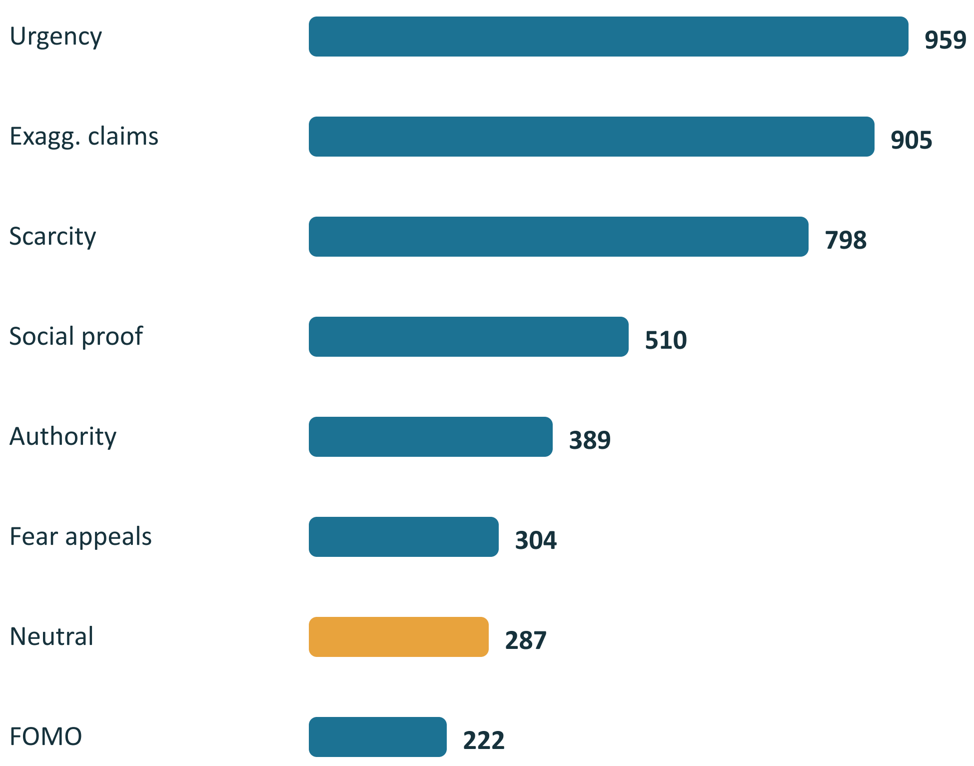
The clean ad — Neutral, no false alarm

Backup: recorded run in `reports/final/demo.mp4`

4,374 ads, 8 classes — rebuilt after we caught our model cheating

- **The shortcut we found:** in the original 7-class corpus (3,230 ads), "limited stock" appeared in **435 of 449** Scarcity examples (96.9%). The model could pass the test by memorizing one phrase.
- **The fix:** merged **857 real e-commerce rows** from the Mathur et al. dark-patterns corpus (Scarcity, Urgency, Social Proof) and added **287 Neutral** examples so "no tactic" became a possible answer.
- **Effect:** top-phrase coverage fell — Scarcity 97.3%→73.4%, Urgency 91.6%→74.0% — forcing the model to learn broader patterns.
- **Protocol:** stratified 70/15/15 train/val/test (657 held-out test ads, untouched during tuning); class-weighted cross-entropy for the 4.3× imbalance.

CLASS DISTRIBUTION (FINAL CORPUS)



Neutral (amber) is new — added with the dark-patterns merge

Fine-tuned DistilBERT, tuned with Hyperopt, shipped small

MODEL

distilbert-base-uncased, fine-tuned for 8-way classification.

128-token training window; longer ads scored in **overlapping sliding windows** with per-class max-pooling — a tactic in an ad's closing seconds still counts.

Metric: macro F1 — with a 4.3× class imbalance, accuracy would hide weak minority-class performance; macro F1 weights all 8 classes equally.

HYPERPARAMETER SEARCH

Hyperopt **TPE** over lr {1e-5…5e-5} × batch {8, 16} × epochs {2–4} × weight decay {0, 0.01, 0.05} — a 72-config space, 3 trials evaluated on validation macro F1.

Winner: lr 5e-5 · batch 16 · 3 epochs · wd 0.0

best 0.886 vs worst trial 0.788 val macro F1, then retrained on the final corpus (val macro F1 0.928)

SHIPPING IT

Weights exported **float32 → float16**: 268 MB → **134 MB**, with zero changed predictions on the held-out check.

Runs on CPU in ~1s per ad — no GPU anywhere in the demo you'll see.

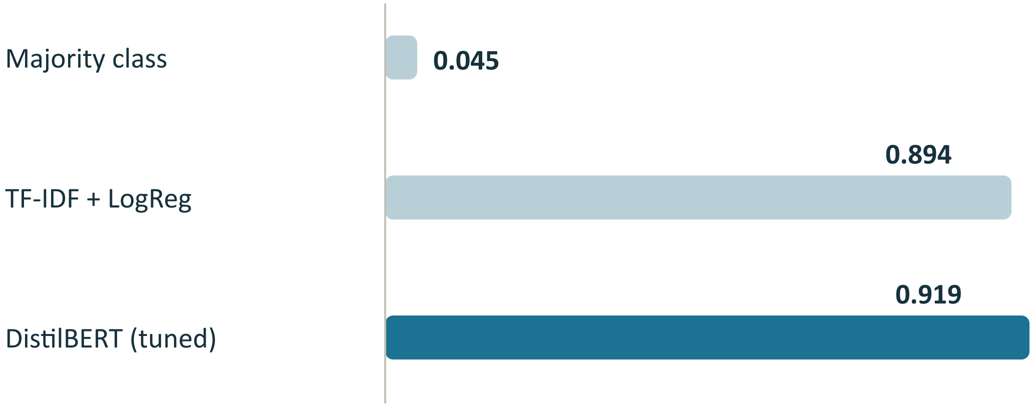
Tokenizer and split logic shared between training and the app — what we evaluate is what we serve.

Reproducibility: the winning configuration and trial table are committed with the repo (`best_parameters.txt · hyperopt_trials.csv`), and the shipped checkpoint was re-evaluated end-to-end on the held-out split while preparing this talk — the reported numbers reproduced exactly.

AREA 3 · NLP METHOD AND EVALUATION — RESULTS

0.919 macro F1 on held-out data — read alongside its baselines

MACRO F1, SAME 657-AD TEST SPLIT



accuracy: 0.219 · 0.901 · 0.929 — identical stratified split, seed 42

7-CLASS → 8-CLASS (AFTER THE DATA REBUILD)

0.896→0.919
macro F1

0.909→0.929
accuracy

0.77→0.88
Social Proof F1

- **Neutral scores a perfect 1.00** — we report that as a warning, not a result: synthetic Neutral rows are trivially separable from tactic ads.
- The 657-ad test set stayed untouched through model selection and tuning.

Honest read: a classical baseline gets close. That is our shortcut story confirmed — formulaic ad phrasing rewards n-grams. DistilBERT's edge (+2.5 macro F1) comes exactly where phrasing varies, and it is the edge that survives on harder, less formulaic ads.

The real world is not our test set

6 GENUINE ADS FROM X / INSTAGRAM / REELS — TEXT EXTRACTION (53 REQUIRED STRINGS)

The OCR bug that started it: an avatar covered "TH", RapidOCR read *"ISWEEKONLY"* — the Urgency trigger never fired and the ad passed as clean.

…BUT CLASSIFICATION BARELY MOVED

1 / 6 correct with OCR text

0 / 6 correct with cleaner VLM text

0 trigger phrases fired in 18 runs

- 4 of 6 were **income-claim ads** — a genre absent from our training data. One scored "Urgency, 98%" with no deadline in the ad: a confident error.
- The 0.60 guard kept the system **silent on 4 of 6** — for a 60+ audience, silence is the right failure mode; a false alarm teaches users to stop trusting the tool.
- Sentence-level probing agrees: out-of-distribution fragments get confident nonsense ("take two capsules daily" → Fear, 99%).

Takeaway: better extraction didn't fix classification — the bottleneck is the training distribution, not the input pipeline. That finding sets our roadmap.

Six users, ages 60–74, four tasks each

5/6
completed the text-ad task without assistance

4/6
completed the image-ad task independently

6/6
identified the main persuasion tactic shown

5/6
rated the plain-language explanation as useful

WHAT IT CHANGED IN THE PRODUCT

- **Kept the extracted-text read-back front and center** — seeing "what the system read" was the most-praised feature; it stays a first-class UI element.
- **Simplifying "model pick" wording** — jargon was the main stumbling block, not the workflow.
- **Rewriting low-confidence copy** so "no strong prediction" can't be read as "verified safe."

EARLIER FIELD RESEARCH SHAPED THE DATA, TOO

Week-8 outreach in ad comment sections: repeated phrases are what raise suspicion —

"As soon as I hear or see 'limited stock', I just scroll away."

That finding drove which trigger phrases and tactics our dataset curated first.

Scope, honestly stated: a six-person, single-session study shows the workflow is learnable and the output understandable — it cannot measure long-term trust or behavior change. A larger, more digitally diverse 60+ cohort is the next validation step.

AREA 5 · ETHICS AND LIMITATIONS

Built to inform, not to judge

DATA RIGHTS

Training data: our collected public ad texts plus the published Mathur et al. dark-patterns research corpus, credited. Labels are educational annotations — never "scam," never a legal claim.

BIAS

The training distribution under-represents whole genres (income-claim ads went 0-for-4). Strong held-out scores are not evidence the model works on every ad — we say so, on stage and in the report.

PRIVACY

Screenshots can contain more than the ad. The RapidOCR path runs fully on-device; VLM and review search leave the device only when API keys are configured — and that boundary must be disclosed to users. Nothing is retained.

WHEN THE MODEL IS WRONG

High-confidence errors happen ("Urgency, 98%", no deadline). Mitigations: the <60% guard prefers silence to false alarms; every finding carries checkable evidence (a quoted phrase or a labeled confidence); the review layer excludes advertiser-owned sources. AdInsight supports judgment — it never replaces it.

"AdInsight gives users a reason to slow down — the decision stays with them."

WHAT WE LEARNED

Three lessons we'd defend

1 · Held-out ≠ real world

0.92 macro F1 in the lab, 1-of-6 on genuine ads. We found it, measured it, and can name the cause: training-distribution coverage.

2 · Explanations need evidence

Users trusted the tool because it quotes the ad, shows what it read, and badges every claim with its source — model pick or phrase found.

3 · For 60+, silence beats a false alarm

One wrong "this is a pressure tactic" teaches a user to ignore the tool. The confidence guard is a design choice, not a limitation.

IF WE KEPT GOING

Multi-label classification head · real-ad training data for uncovered genres (income claims) · a larger usability cohort · hardening the review layer's retrieval.

Thank you — questions?

github.com/cpedrett-umd/Group-6-Final-Project · Chris Pedretti · Shashank Ashoka · Ciara Cameron · Jonathan Kim